Geodesic motion in weak gravity

As all particle orbits are planar in the Schwarzschild metric, we can write it in the equatorial plane, $\theta = \dfrac{\pi}{2}$, as

$$\mathrm{d}s^2 = c^2\sigma(r)\ \mathrm{d}t^2 - \sigma^{-1}(r)\ \mathrm{d}r^2 - r^2\mathrm{d}\phi^2; \qquad \sigma(r) \equiv 1 - \frac{r_\mathrm{s}}{r}\,, \qquad r_s \equiv \frac{2GM}{c^2}\,.$$

Geodesics in this metric describe motion of particles or light in gravitational field of a spherically-symmetric mass distribution of radius R. In this sheet, we consider only week gravity when $r \geqslant R \gg r_\mathrm{s}$.

1. For a particle's radial fall from rest near the Earth surface, we have $h = h_0 - \frac{1}{2}gt^2$ and $v = -gt$. Derive these (trivial) Newtonian results geometrically from the Schwarzschild metric.

2. Three observers have identical precise atomic clocks synchronised at the Earth surface. The first throws his clock vertically. The second travels up and down in the vertical direction at a constant speed, keeping her clock in hands. She reaches the same maximal height as the first observer's clock, and returns at the same moment when the first observer catches up his clock. This happens in time $T_3 = T = 20\,\mathrm{s}$ as measured on the clock of the third, stationary observer. Assuming that there is no air resistance (which is totally unrealistic but could be possible, say, on the Moon), find differences in the times $T_{1,2}$ registered on the clocks of the first and second observers from the time T measured by the third. Get your answer analytically and numerically, using the Earth radius $R \approx 6.4 \times 10^6\mathrm{m}$, $g \approx 9.8\,\mathrm{m/s}^2$, and $c \approx 3.0 \times 10^8\,\mathrm{m/s}$.

3. Show that any ***circular*** orbit of a particle in the Schwarzschild metric ***exactly*** obeys Kepler's 3d law, $\omega^2 r^3 = GM$, where $\omega = \mathrm{d}\phi/\,\mathrm{d}t$ is a coordinate angular velocity. Comment on this result.

4. A satellite is in a circular orbit of radius $2R$ around the Earth. An atomic clock on the satellite is synchronised with an identical clock on earth and found after one revolution to be some time ***ahead.*** Find the radius of another satellite orbit in terms of the earth radius R where the clock would be the same time ***behind*** the clock on earth after one revolution.

5. The GPS configuration consists of a network of 24 to 30 satellites, each having two rotation a day around the Earth.

   (a) Assuming that orbit of each satellite is circular, find its altitude.

   (b) Use this to estimate roughly the atomic-clock accuracy required for positioning an object with accuracy of $1\mathrm{m}$.

   (c) Calculate a time difference, which is due to the gravitational and kinematic time dilation, between the precise atomic clocks on the satellite and on the Earth accumulated over one rotation.

   (d) Suggest in what way this time difference can be compensated to achieve the required accuracy. (You can google this but you might come with a better idea).

   Hint: For a satellite with the angular velocity ω on a circular orbit of radius r, use Kepler's law, $\omega^2 r^3 = GM$, and the first integrals of the Schwarzschild metric for equatorial motion.
   For numerical estimates use the values of the Earth radius, $R_0 \approx 6.4{\times}10^3$ km, and of the free fall acceleration, $g \approx 9.8$ m/s^2.

6. Using the Schwarzschild geometric Lagrangian, find the total energy of a satellite of rest mass m_0 in a circular orbit of radius r around a spherical planet of mass M. Find the non-relativistic limit of your result and show that it coincides with the classical $\mathcal{E} = T + U$, which you should simplify using Kepler's 3rd law.

7. A non-relativistic equation for Kepler's orbit in a gravitational field created by a star of mass M is

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\phi^2} + u = \frac{GM}{J^2}\,,$$

where $u \equiv 1/r$, is the orbital variable with $(r,\, \phi)$ being polar coordinates in the plane of motion, J is the angular momentum of a planet moving in the field.

   (a) Derive this equation geometrically, using first integrals for the Lagrangian corresponding to the 'pseudo-Newtonian' metric, $\mathsf{g}_{\mu\nu} = \mathrm{diag}\,(\sigma,\, -1,\, -1,\, -1)$.

   (b) Find the value of J for a circular orbit of radius R.

   You should assume that the star is spherical, with radius $R_0 \ll R$ and consider the planet as a material point (i.e. neglect its size, internal rotation, etc).

8. Derive the equation for Kepler's orbit directly from Newton's law of gravity.

   [*Hint: The orbital Kepler's equation is general but in your derivation you may restrict considerations to circular motion and use polar coordinates.*]

9. The deflection of light could be predicted on the basis of Newtonian theory, with the only assumption that *light is heavy* ($m = \mathcal{E}/c^2 = \hbar\omega/c^2$ – obviously, not a rest mass!). Applying to light the equation describing orbital motion of *massive* particles, or otherwise, show that the total deflection in this case would be $2\delta\phi = r_\mathrm{s}/R$, *i.e.* twice smaller than predicted by GR. Can you explain the mathematical origin of an extra factor of 2?

10. A projectile is fired from the Earth surface at speed $v \ll c$ at angle α to the horizon. Describe its motion using a general solution to the Kepler's orbit equation. In particular,

   (a) find escape velocity v_0 (a minimal velocity at which orbit is hyperbolic)

   (b) find the conditions of setting a *projectile* to a periodic elliptic orbit around the Earth

   (c) explain how such a description goes over at $v \ll v_0$ to the standard (high-school) description of a parabolic motion in an uniform gravitational field near a 'flat' earth surface.